

IFDP Quick Reference Card

The Investigation-First Development Protocol

A three-phase, evidence-driven approach to fixing production issues.

Quick Overview

Phase	Duration	Focus	Deliverable
A: Investigation	2-4 hours	Root cause analysis with data	Hypothesis + approval
B: Implementation	2-8 hours	Surgical fix + observability	Code + tests deployed
C: Verification	24-48 hours	Production monitoring	GO/NO-GO decision

Phase A: Investigation & Diagnosis

Rule: NO CODE CHANGES

- 1. **Flow Mapping** → Trace logic with file paths & line numbers
- 2. **Evidence Gathering** → Run database queries, analyze logs
- 3. **Code Review** → Inspect affected modules & configs
- 4. **Hypothesis Formation** → State root cause + 3-5 fix options

Stop Condition:

Phase [X]A investigation complete. Root cause: [Finding]. Awaiting approval to proceed to Phase [X]B.

Phase B: Implementation

Focus: Surgical fixes + observability

- 1. **Surgical Fixes** → Only approved changes, no scope creep
- 2. **Observability First** → Add logging BEFORE changing behavior
- 3. **Testing** → Unit tests + manual validation (100% pass)
- 4. **Deployment** → Deploy & verify clean logs

Logging Standard:

```
{
  "event_type": "fix_checkpoint",
  "context": { "city": "...", "slot": "...", "provider": "..." },
  "message": "...",
  "error_details": { "http_status": "...", "error_message": "..." }
}
```

Stop Condition:

Phase [X]B implementation complete. Deployed successfully. Ready for Phase [X]C.

Phase C: Production Verification

Duration: 24-48 hours

1. **Monitoring** → Observe production without changes
2. **Metric Comparison** → Compare before/after data
3. **Final Audit** → Check for regressions

Stop Condition:

Phase [X]C verification complete. Final verdict: [GO/NO-GO with reasoning].

Key Principles

- ✓ **Observability Before Action** — Add logging before changing behavior
- ✓ **Evidence-Driven** — Base decisions on data, not intuition
- ✓ **Independent Revertability** — Each phase rolls back cleanly
- ✓ **Minimal Changes** — Surgical fixes only, no improvements
- ✓ **Standardized Logging** — Consistent format across all phases

When to Use IFDP

- ✓ Production outages
- ✓ Recurring issues
- ✓ Complex system failures
- ✓ High-impact bugs
- ✓ Pattern-based issues
- ✗ Simple typos
- ✗ Obvious one-liner fixes
- ✗ Issues with known solutions
- ✗ Low-impact bugs (use IFDP lite)

Real-World Results

Issue	Phase A Finding	Result
Branding Bug	Hardcoded banned fragments in templates	Block rate: 25-30% → <2%
Rendering Variety	Silent API failure causing fall-back	Diversity improved + no regressions

Phase Checklist

Phase A

- ☐ Flow diagram created
- ☐ Evidence gathered (queries, logs, metrics)
- ☐ Root cause hypothesis formed
- ☐ 3-5 fix options identified
- ☐ Approval obtained

Phase B

- ☐ Approved fix implemented
- ☐ Observability logging added
- ☐ Tests passing 100%
- ☐ Deployment clean
- ☐ Rollback plan documented

Phase C

- ☐ 24-48 hours monitored
 - ☐ Before/after metrics compared
 - ☐ No regressions detected
 - ☐ Final verdict: GO or NO-GO
-

Commands (for Git Integration)

```
# Phase A: Investigation findings
git commit -m "Phase 5A investigation: root cause identified - hardcoded fragments"

# Phase B: Implementation
git commit -m "Phase 5B implementation: rewrite templates + consolidate title-builder"

# Phase C: Verification
git commit -m "Phase 5C verification: GO - block rate 25% → <2%"
```

For AI Assistants

When requesting help:

- “Help me with Phase 3A investigation for [issue]”
- “Let me implement the Phase 3B fix based on my Phase 3A findings”
- “Help me verify the Phase 3C metrics for this production fix”

Reference: See `IFDP_METHODOLOGY.md` for full details

Lovable/Cursor: See `LOVABLE_INSTRUCTIONS.md`